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Fake News Detection using Deep Learning

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Bachelor of Computer Science (Data Science)

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Abstract

With the rise of social networking the spread and volume of fake news has dramatically increased, to the point where fact checking every post is unrealistic. In attempts to solve this problem, researchers have proposed methods employing machine learning and deep learning to flag fake news. However, these methods are often limited by lack of applicability as they fail to generalise beyond the datasets used. Furthermore, many of these methods employ black-box models, which reduces trust in the model's predictions. In order to address these issues, we proposed a model based on adversarially fine-tuning a Bidirectional Encoder Representations from Transformers (BERT) model to increase generalisation, and the use of Local Interpretable Model-Agnostic Explanations (LIME) to explain the otherwise black-box BERT model. Results indicated that the proposed model did not achieve its aims of increased generalisation. However the LIME implementation did result in it being able to explain the models somewhat, but more importantly it demonstrated a behavioral difference in between BERT and the proposed model.

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List of Abbreviations

Abbreviation	Full Form
AFND	Arabic Fake News Dataset
AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
Bi-GRU	Bidirectional Gated Recurrent Unit
Bi-LSTM	Bidirectional Long Short-Term Memory
BiGRU	Bidirectional Gated Recurrent Unit
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CNN-LSTM	Convolutional Neural Network - Long Short-Term Memory
Conv-FFD	Convolutional Feed-Forward
DAGA-NN	Domain-Adversarial & Graph Attention Neural Network
DFT	Discrete Fourier Transform
DistilBERT	Distilled BERT
DL	Deep Learning
Doc2Vec	Document to Vector
EANN	Event Adversarial Neural Network
EBGCN	Edged-enhanced Bayesian Graph Convolution Network
ELA	Error Level Analysis
FGM-FRAT	Feature Regularization-based Adversarial Fine-tuning
GBERT	Generative Bidirectional Encoder Representations from Transformers
GBCA	Graph Convolution Network and BERT combined with Co-Attention
GCN	Graph Convolution Network
GloVe	Global Vectors for Word Representations
GPT	Generative Pre-Trained Transformer

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List of Abbreviations (continued)

Abbreviation	Full Form
GRU	Gated Recurrent Unit
GRU-RNN	Gated Recurrent Unit - Recurrent Neural Network
ID	Identification
IFND	Indian Fake News Dataset
LLM	Large Language Model
LIME	Local Interpretable Model-Agnostic Explanations
LRP	Layer-wise Relevance Propagation
LSTM	Long Short-Term Memory
MCNN	Multimodal Consistency Neural Network
ML	Machine Learning
MPFN	Multimodal Progressive Fusion Approach
OPCNN-FAKE	Optimized Convolutional Neural Network to detect fake news
PMI	Pointwise Mutual Information
POS	Part of Speech
ReLU	Rectified Linear Unit
ResNet50	Residual Network 50
SAT	Situation Awareness-based Agent Transparency
SDL	Sequential Deep Learning
SVM	Support Vector Machine
SVM-TK	Support Vector Machine - Tree Kernel
SwinT	Swin Transformer
TD-IDF	Term Frequency - Inverse Document Frequency
TFNDS	Toxic Fake News Detection System
VGG19	Visual Geometry Group 19
Word2Vec	Word to Vector
XAI	Explainable Artificial Intelligence
XGBoost	eXtreme Gradient Boosting

1 Introduction

1.1 Problem Statement

Fake news threatens personal, organisational and institutions by creating distrust and damaging reputations. Fake news can be particularly damaging as even if it can be refuted, it may still cause lasting damage to reputations. With the widespread use of social media, the proliferation of fake news has greatly increased both in terms of speed and quantity. This has led to the sheer volume of fake news (and real news) overwhelming human fact-checkers and preventing them from manually verifying all content.

But what is fake news? According to Baptista and Gradim (2022) the definition of fake news is generally agreed to be either completely or partially false information, with varying levels of agreement for other additional criteria. Such as the intent to deceive, and if only content in a news format should be counted. For the purpose of this research, the working definition of fake news defined by Baptista and Gradim (2022) will be used:

> a type of online disinformation (1), with (2) misleading and/or false statements that may or may not be associated with real events, (3) intentionally created to mislead and/or manipulate a public (4) specific or imagined, (5) through the appearance of a news format with an opportunistic structure (title, image, content) to attract the reader's attention, in order to obtain more clicks and shares and, therefore, greater advertising revenue and/or ideological gain Baptista and Gradim (2022)

In a recent (as of writing) example of fake news causing adverse impact, Savage (2025) wrote for The Guardian about YouTube channels spreading fake news about UK politics in their videos, accruing close to 1.2 billion views in total. While some forms of news which are factually fake are usually harmless (parody, satire, etc.) fake news, with the intent or deceiving those who consume it is dangerous, it can widen division, spread prejudice and cause distrust in institutions are among just a few examples. And while fake news may sound innocuous enough, 'It's just fake news, ignore it', it can lead to serious consequences, including injury and death. In one such example, during the COVID-19 vaccination campaign in Taiwan an adverse correlation between prevalence of fake news and vaccination rates were found (Chen et al., 2022).

As a result, the development of automated fake news detection systems have been explored. But limitations related to generalisation where detection systems perform poorly on data which did not appear in its training sets (Qin & Zhang, 2024) limit their applicability. Additionally, current methods such as Machine Learning (ML) approaches, fail to capture the semantic data meaningfully, reducing its detection accuracy. Furthermore, many approaches which employ Deep Learning (DL) are often black-boxes, which results in lowered trust towards these systems (Szczepański et al., 2021).

This highlights the need for new approaches which are capable of detecting fake news by analysing the semantic information in the text thoroughly, having greater generalisation and be more transparent (explainable) when making its classifications.

1.2 Research Objectives

The objectives of this project are:

- Review on methods to detect fake news
- Develop a fake news detection using deep learning
- Evaluate the performance of model in detection fake news

1.3 Research Scope

This project aims to build a deep learning model for fake news detection by first conducting a thorough literature review on related works in order to construct a knowledge base of which will be used to construct a model for detecting fake news.

After the literature review, models to detect fake news will be developed using deep learning techniques. This process will involve gathering datasets, analysing and understanding the data and pre-processing datasets before constructing the architecture of the models and then training the various models.

And lastly, after training the models. The models will be tested on the test sets which are subsets of the datasets used in order to determine their performance.

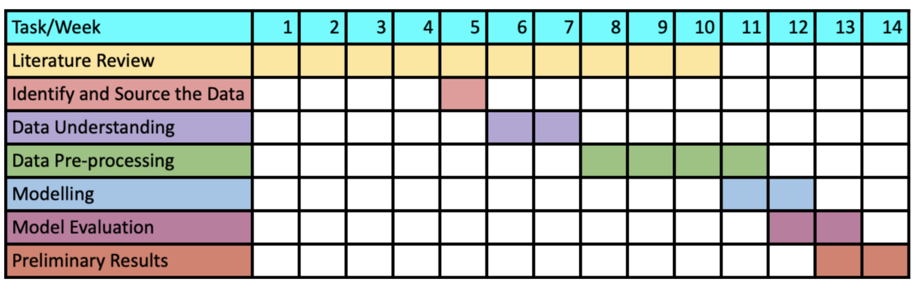

Figure 1.1: Project Timeline Gantt Chart for MMU Term 2530

2 Literature Review

2.1 Deep Learning Methods and Other Modelling Techniques

2.1.1 Convolutional Neural Networks

In research conducted by Q. Zhang et al. (2023), which detected fake news on Chinese social media networks using an augmented Convolutional Neural Network (CNN), Convolutional Feed-Forward or Conv-FFD. In order to process the Chinese characters, the researchers used a three-step process starting with one-hot-encoding the over 4000 Chinese characters, then feeding this data in an embedding process which converts the one-hot-encoded data into vector representations of each character, culminating in iterating through all the characters to get the vector representation of each one (essentially mapping the characters to the vector embedding). As for the feature modelling process (the process of understanding the relationship between characters/words), the researchers used the CNN model which is able to learn hidden features from conducting convolution operations. They used a 'sliding window' in order to create sentence-level embedding. Then in order to classify the text as real or fake, a softmax activation function is used in order to predict the class of the input text. The model achieved an F1-score of 0.8702 - 0.8855, indicating it is capable of predicting whether a given body of (Chinese) text is real or fake.

Research conducted by Saleh et al. (2021) also proposed a CNN model, albeit a modified one they called the "optimized Convolutional Neural Network to detect fake news" or OPCNN-FAKE. Aside from the proposed models, the authors also built machine learning models and other deep learning models like LSTMs The authors used four datasets, Fake News Detection (from Kaggle), FakeNewsNet, FA-KES5, and the ISOT dataset. The datasets were cleaned by converting all characters to lower-case, removing URLs, removing special symbols, removing stop words, tokenisation and stemming (reduction to root word). After the data was cleaned, two methodologies were used for feature generation, one for training the machine learning models and one for the deep learning models. For machine learning models, the feature generation used two techniques, N-grams (up to n = 4) and Term Frequency - Inverse Document Frequency (TD-IDF). While for the deep learning models, feature generation was done via word embeddings (the authors classified this as the "Embedding Layer" or the first layer of the proposed OPCNN-FAKE model), specifically the Global Vectors for Word Representations (GloVe) which converted the cleaned textual data into dense vectors (more specifically they used GloVe6B.zip). The preprocessed datasets were then split 80-20 into training and test sets respectively and the training set was used to train the machine learning and deep learning models, including the proposed OPCNN-FAKE model. For the deep learning models, a technique called "Hyperopt Optimisation" was used to adapt the various model parameters including epoch, dropout and batch size, among others. The proposed OPCNN-FAKE model was configured with six layers, the embedding layer, dropout layer, convolutional layer, pooling layer, flatten layer and output layer respectively. Without diving too deeply into the architecture, the embedding layer was previously explained; the dropout layer randomly deactivates some neurons to mitigate overfitting; the convolutional layer uses a convolution filter and feature map to identify the data patterns and high and low level features while employing a ReLu activation function; the pooling layer reduces the number of features in the feature map so as to only retain the most important features; the flatten layer flattens the feature map into a one-dimensional array; the output layer takes the one dimensional array from the flatten layer and classifies it as fake (0) or true (1). The models were then evaluated using cross-validation and the OPCNN-FAKE model achieved the best F1-score across all four datasets used, with the proposed model achieving an F1-score of 0.959 on the Kaggle dataset and a perfect F1-score of 0.9999 on the ISOT dataset.

2.1.2 Hybrid Approaches

There has also been research that looked into hybrid approaches, Hashmi et al. (2024) proposed a hybrid deep learning model combining Long Short Term Memory (LSTM) and CNN. Using publicly available datasets (WELFake, FakeNewsNet and FakeNewsPrediction), the authors first pre-processed the data using lemmatisation (reduction of words to root form), de-capitalisation, tokenisation, among others. To process the cleaned data, the authors used

FastText, both its supervised and unsupervised forms to create word embeddings. These words embeddings, generated from both the unsupervised and supervised FastText models were then used to train separate models, including ensemble (stacked) machine learning models, transformer models and the CNN-LSTM hybrid mentioned earlier. Evaluation of the models using test subsets of each of the datasets used showed that those trained on unsupervised FastText embeddings performed worse than those trained on supervised FastText embeddings, with F1-scores between 0.83 - 0.97 with significant variation between the datasets it was tested on. Those trained on supervised FastText embeddings were evaluated on subsets of the datasets used (test sets) and had F1-scores of between 0.91 - 0.99. The CNN-LSTM model performed better than all other models (stacked machine learning models and transformer models), with high F1-scores of between 0.97 and 0.99 across all datasets.

While most research seems to have been done on detecting fake news in English, research conducted by E.Almandouh et al. (2024) detected fake news in Arabic. The authors used two datasets, Arabic Fake News Dataset (AFND) and their own dataset scraped from Twitter. The authors explored multiple types of models for detecting fake news including machine learning and common deep learning models but they also proposed a hybrid model called Bidirectional Long Short Term Memory – Bidirectional Gated Recurrent Unit (Bi-LSTM – Bi-GRU). This proposed model is structured sequentially, but it can also deal with information bi-directionally and it attempts to leverage the strengths of Bi-LSTMS and Bi-GRUs. Bi-LSTMs and LSTMs in general are excellent at retaining information and addresses the vanishing gradient problem; while GRUs are able to manage shorter dependencies and are computationally efficient. For preprocessing, the authors first cleaned the data by removing all punctuation, Unicode characters, names, URLs, stop word removal, tokenisation and other standard techniques. The cleaned data was then features were extracted using both supervised and unsupervised FastText which were then used to train models. The Bi-LSTM – Bi-GRU performed the best among all of the models tested, achieving F1-scores of between 0.98 and 0.99 depending on the dataset.

Dhiman et al. (2024) proposed a new deep learning model for predicting fake news called "Generative Bidirectional Encoder Representations from Transformers" or GBERT, a hybrid of Generative Pre-Trained Transformer (GPT) and Bidirection Encoder Representations from Transformers (BERT). The authors used two benchmark Indian datasets: The Indian Fake News Dataset (IFND) and FakeNewsIndia Dataset. The standard cleaning and normalisation of the textual data was applied including: decapitalisation, removal of URLs, names, punctuation stop words, numbers and emojis, and the application of stemming and lemmitisation. Now for the feature extraction the cleaned data is fed into two Large Language Model (LLM) components, BERT and GPT. BERT captures word embeddings, with context from every token input into it and outputs a summary of the context inside the input sequence. While GPT captures the global dependencies and semantic coherence in the input data and outputs the last hidden state of the final token. Finally the output from BERT and GPT (the features) are merged together in a concatenation layer before being passed to the dense layer for the final binary classification using a Sigmoid function. The results showed that the proposed GBERT model performed well, with a F1-score of 0.9623. It is worth noting, however, that XGBoost performed well in terms of accuracy (F1-score was not provided for XGBoost) with a score of 93.42% while the proposed GBERT model has a score of 95.30%.

2.1.3 Deep Ensemble Approaches

Other than approaches with CNNs, LSTMs and machine learning models, there have also been attempts to construct deep ensemble models composed of deep learning models arranged in an ensemble architecture. Research conducted by (Ali et al., 2022) proposed a new approach for detecting fake news, by using Sequential deep learning (SDL) models as the base models in an ensemble stacking architecture with a multilevel perceptron classifier at the top. The authors used two datasets: the LIAR and ISOT datasets. The data from each dataset was preprocessed with tokenisation, lemmatisation among other standard techniques for working with textual data. Unlike with other similar works, the authors chose to do multi-class classification because fake news is usually mixed with true news. For the SDLs they are composed of dense and dropout layers. The ReLu activation function is used in all dense layers except the final output layer where a Sigmoid function was used. The ensemble model was evaluated with the test subset of each dataset used, for the LIAR dataset the model achieved F1-scores in a range from 0.48 to 0.82 for the various classes. As for the ISOT dataset, since the dataset only had two classes (real and fake), the model only had to classify true and false, the binary classification was able to achieve a 1.00 accuracy and F1-score using the 2nd stage classifier.

2.1.4 Machine Learning Approaches

Research conducted by (Khanam et al., 2021) explored the use of machine learning models such as XGBoost, Support Vector Machine (SVM), Naive Bayes, among others. The authors used the LIAR dataset which they then preprocessed by removing noise such as identification (ID), dots, commas and questions as well as stemming (reduction of a word to root form) and removing any suffixes. They then use Part of Speech (POS) to convert the text into tokens and values. They then extracted features by extracting lexical features such as word count, sections of speech etc. and then used the Term Frequency-Inverse Document Frequency (TD-IDF) vectoriser function from the publicly available library sklearn (also known as scikit-learn) to extract unigram and bigram features. The results showed that the accuracy of the models was limited, especially in comparison to approaches using deep learning with the best performing model being the XGBoost model with an accuracy of 0.73.

2.1.5 Multimodal Approaches

In research conducted by Jing et al. (2023) the researchers employed a multimodal approach instead of a purely textual approach to detection. To this end, the authors proposed the novel, end to end Multimodal Progressive Fusion Approach or MPFN. The authors used two datasets, a Twitter dataset and a Weibo dataset. The MPFN model has three components, a text feature extractor, visual feature extractor and the progressive multimodal feature fusion process. Starting with the text feature extractor, the authors used a pre-trained Bidirectional Encoder Representations from Transformers (BERT) in order to extract the features from the text, this BERT model returned text features with 768 dimensions. The visual feature extractor has two major subcomponents, the first seeking to obtain spatial semantic information and the second seeking to obtain features which can indicate whether an image has been tampered with. The first component starts off by setting all images to 224 x 224 x 3 (width, height, colour), then to extract features the authors used a pre-trained Swin Transformer (SwinT) model to extract the spatial semantic information of the image. This is done by splitting the images into 4 x 4 x 4 non-overlapping patches and each patch is passed through four layers in order to obtain a hierarchical representation. As for the second subcomponent, the authors extracted frequency domain features using Discrete Fourier Transform (DFT) and Visual Geometry Group 19 (VGG19). DFT was used to convert the image from the spatial domain to the frequency domain and the output is inputted into the VGG19 model which outputs deep semantic features; however in order to obtain a hierarchical data structure, the outputs from the 2nd, 4th, 8th and 16th convolutional layers in the VGG19 model is extracted for use in the fusion model later. The outputs from the text extractor, and the outputs from the visual feature extractor are then combined in a progressive fusion approach where the shallower layers in the hierarchical data structure are formed first and so on for deeper layers (there are 4 layers total). Mlp mixer was used to fuse feature information in a finer way and enhanced the relationships between modes. The final fusion features are then used for classification by a fully connected layer using a softmax function; additionally the model uses a cross-entropy loss function for calculating the difference between the true label and predicted probability. The results showed that this approach achieved an F1-score of 0.889 - 0.876 for both the Weibo and Twitter datasets for both classes with the exception being a 0.740 score achieved by the model when classifying real news on Twitter.

In research conducted by Xue et al. (2021), the authors proposed a novel 5-module model "Multimodal Consistency Neural Network" or MCNN for multimodal (image, text) fake news detection as depicted in Figure **??**. The text feature extraction model uses a BERT pre-trained model to encode text and then uses Bidirection Gated Recurrent Unit (BiGRU) to extract temporal features and convert features into a sequence that is suitable for integration with image features. For the visual semantic feature extraction module, a pre-trained ResNet50 model is used to encode the image which is then fed into an attention mechanism to highlight emotional regions of the image followed by the use of BiGRU to convert the features into a visual semantic sequence. The visual tampering feature extraction model, unlike the previously described module seeks to extract features at a physical level in order to detect malicious tampering with the image, it uses the Error level Analysis (ELA) algorithm to process input images before using ResNet50 to extract features. The similarity measurement module maps the text and image semantic representations into one space, followed by the calculation of the cosine similarity; this is done to address the issue of image-text mismatch. This is finally followed by the multimodal fusion module which takes input from the similarity measurement module and the visual tampering feature extraction module and uses an attention mechanism to give the combined features weights before using a softmax classifier to classify the image. The model was tested against various baseline models, both single and multimodal models were used. Four distinct datasets were used and the model performed the best in terms of F1-score across all datasets save one, losing out to a BERT-MVNN hybrid (note: there appears to be an error in the paper where the authors failed to notice that the BERT-MVNN model outperformed the proposed model in the results table). The proposed model's F1-score ranged from 0.968 - 0.831.

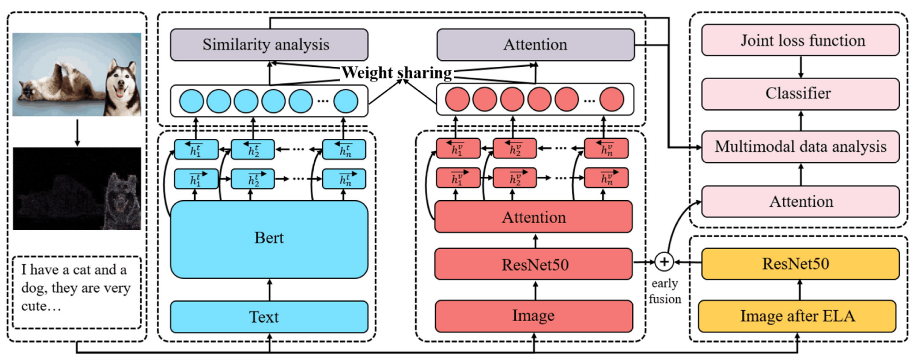

Figure 2.1: Architecture of the Proposed MCNN Model. From "Detecting fake news by exploring the consistency of multimodal data", by Xue et al., 2021, *Information Processing and Management.*, 58(5), https://doi.org/10.1016/j.ipm.2021.102610. Copyright 2024 held by 2021 Elsevier Ltd.

(Yuan et al., 2021) proposed a new model for detecting fake news, the "Domain-Adversarial & Graph Attention Neural Network" or DAGA-NN. This approach is multimodal and takes both image and text as input. The proposed model was designed in order to address the challenge machine learning approaches faced when attempting to classify cross-domain news samples that did not (or had few instances of) appear in its training data. The architecture is depicted in Figure **??**. Starting with the feature extractor module, the text component is first cleaned and tokenised before being passed to a BERT model to gain vector representations which are then passed to the Bi-LSTM to capture information about the text and temporal features. As for the image feature extraction, the input image is fed to a pre-trained VGG19 model and the output is passed to a fully connected neural network to gain the feature representation of the image. The text and image features are then fused together to form a single representation of the image and text. The fused feature representation is then fed to the domain discriminator whose goal it is to determine the domain information of the input news. The model is designed in such a way that the domain discriminator is put in competition with the feature-extractor module, forcing the module to output features which are not (or at least minimally) domain-specific, which helps the classifier train on features not dependent on domain and increasing its cross-domain classification accuracy. The domain discriminator is a two-layer neural network which outputs a vector of k-dimensions representing its probability of belonging to each of the k domains. Moving on to the graph-attention-based-classifier, it first takes news articles and creates a graph of them. Edges are drawn between nodes if their cosine similarity is high. Once the graph is built, the graph attention layer learns the importance of each node's neighbours when classifying an input article. This is critical as during this phase, the graph attention layer will strengthen connections between nodes (articles) which are similar semantically and have the same veracity label (true or false) and vice versa (weaken connections if veracity label is opposing). This establishes clear boundaries between real and fake news, even if they are semantically similar. The DAGA-NN model was tested against two datasets, the Twitter and Weibo datasets and managed to outperform state-of-the-art models. The DAGA-NN model was able to achieve macro F1-scores of 0.8828 and 0.9522 for the Twitter and Weibo datasets respectively. The best baseline model the DAGA-NN model was tested against was the Event Adversarial Neural Network or EANN and it outperformed it by 0.0890 - 0.0112 in terms of macro F1-score.

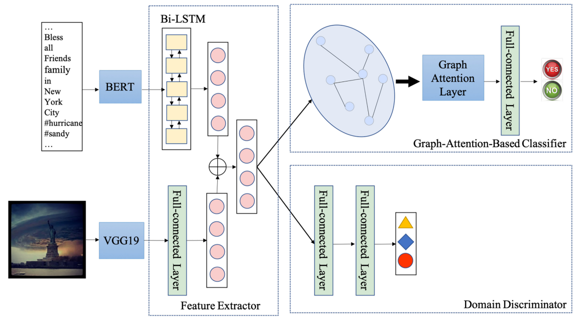

Figure 2.2: Architecture of the DAGA-NN Model. From "Improving fake news detection with domain-adversarial and graph-attention neural network" by Yuan et al., 2021, *Decision Support Systems*, 151, p. 3 (https://doi.org/10.1016/j.dss.2021.113633). Copyright 2021 Elsevier B.V.

2.1.6 Language Model Approaches

The rise of Large Language Models (LLMs) have complicated detection of fake news, to this end (Wu et al., 2024) proposed a new model "SheepDog" leveraging LLMs to improve detection. The model utilises two different models, an LLM for news re-framing and a fine-tuned language model for classification. Firstly, SheepDog utilises an LLM to transform news content into different styles so as to remove stylistic with varying prompts in order to transform a dataset into one which there is minimal stylistic differences between real and fake news. Then the LLM is further prompted with a set of questions to which the LLMs responses are used as pseudo-labels. Moving on to the language model stage, the language model is fine-tuned using labelled datasets processed by the LLMs. The language model will take input from the LLM stage and output the prediction (true or false news). SheepDog is designed to be able to work with any language model and as such the authors tested the SheepDog framework with several language models. For evaluation the authors tested SheepDog against LLM-empowered style attacks, unperturbed (original) articles, and adaptability with different language models. For the baseline models SheepDog was compared against, the authors used three main groups of models: text-based fake news detectors, fine-tuned language models and LLMs. And the researchers used three datasets: PolitiFact, GossipCop and LUN. Sheepdog proved to be resilient against LLM-empowered style attacks and its performance was superior to all baseline models. When tested against unperturbed articles, SheepDog had comparable or superior performance relative to the baseline models with F1-scores of between 0.9304 - 0.7575. As for adaptability with various language models, the SheepDog framework was able to improve the performance of these language models. The best language model which was used as a backbone in the SheepDog framework depended on which dataset they were tested against but all language models integrated into the SheepDog framework performed better than its unintegrated counterpart with F1-scores of between 0.7354 - 0.8563 compared to the (unintegrated; lone) language models which had F1-scores between 0.5247 - 0.7617.

2.1.7 Explainable Models

Research by Szczepański et al. (2021) proposed a novel approach which sought to make the model's decision-making explainable (explainable AI), the reason being that having a black-box system determine the veracity of something doesn't inspire confidence, something an explainable AI model could mitigate. The researchers utilised DistilBERT which is a lighter variant of BERT, combined with an LSTM layer, a pooling layer, dense feed forward layer and finally the output layer. They then attached an explainability module to the black-box classifier. This module is composed of two post-hoc explainable AI techniques, Local Interpretable Model-Agnostic Explanations (LIME) and Anchors. Post-hoc refers to techniques which attempt to give some level of transparency to an otherwise black-box model. LIME works by sampling the model in the vicinity of a prediction (imagine having an unknown 'X' on a spot on a graph and then picking spots around it to approximate the 'X''s coordinates) and the response to the sampling is used to train a linear model that approximates the model's prediction that can be interpreted. While Anchors refer to an algorithm that applies 'rules' to local predictions, the rules being 'if-then' statements which should ensure that any prediction will remain the same wherever the rule holds. The model was able to achieve 0.98 F1-score for both positive and negative classes. The researchers were able to conclude from the results that it is feasible to use methods such as LIME and Anchors to explain the models behaviour. Although there was some overlap between LIME and the

Anchors when explaining the model's behaviour, the results showed it is possible to use multiple types of these methods can be used to derive explanations.

Chien et al. (2022) proposed an explainable AI approach, XFlag, to fake news detection which sought to increase the transparency and trust of black-box AI models. The authors used the Weibo dataset, which was processed to extract content, user and sentiment features. Content features were extracted with Doc2Vec which generates word embeddings from the content, similar to BERT, albeit Doc2Vec is an older model. User features were extracted from the user's profile, such as followers, location, etc.. Sentiment features were extracted with the Google Cloud Natural Language API which returned the emotional, "sentimental" data from the content. These features are then passed to fully connected layers before being passed to the LSTM in order to standardise the input to the LSTM. XFlag was able to perform comparably to state of the art models, with an F1-score of 0.938. As for the explainability, the authors used the Layer-wise Relevance Propagation (LRP) algorithm which uses reverse propagation, essentially start with the model output and going in the reverse direction through the model's various layers to determine which neurons had the most impact on the result. The output of the LRP would be the model's (being tested) input shape but with the relevance score for each neuron, this is known as the explanation vector. LRP was applied to the XFlag model after the model was trained, and the output explanation vectors are then passed to Situation Awareness-based Agent Transparency (SAT) framework, specifically the SAT-3 (the researchers used other SAT versions from SAT 1 - 3 which each iteration having more insightful explanations, SAT-3 was most preferred by users) which takes this raw data and processes it in order to generate visualisations, textual annotations and the model's uncertainty so that the user can see how the model came to the conclusion that it did. Users were tested on their trust of the XFlag model with various SAT versions, the lowest being SAT-1 which, similar to a model with no explainability, output the result without deeper explanation. The results were that users reported highest satisfaction and trust as the SAT level and correspondingly, the depth of the explanations of the model's prediction, increased, the highest being SAT-3.

Research by M. Ahmed et al. (2022) explored the use of explainable AI (XAI) for detection of toxic COVID-19 fake news, proposing the Toxic Fake News

Detection System (TFNDS) which uses Bidirectional Long Short Term Memory (BiLSTM) for classification and similar to Chien et al. (2022) the authors used LIME to explain the black-box model (the BiLSTM). The authors used two datasets which they subsequently merged together, the data was then preprocessed with removal of stop words, removal of emojis, removal of repetitive posts, lemmatisation and etc.. This data is then passed to a scikit-learn (also known as sklearn) pipeline where lists of words are indexed and turned into lists of indices and the length of this list is standardised by a transformer to a maximum length of 100. This is then passed to the model which is composed of 128 BiLSTM hidden units followed by 3 dense layers with a Sigmoid output layer for the final classification. As for the explainable section, LIME augments the data by removing features to see which features (words) was most important to the output and this information is visualised by highlighting the words. On the primary dataset used, the model was able to achieve an F1-score of 0.9425 and when tested on an additional dataset for verification (WNUT-2020) it achieved an F1-score of 0.894 which outperformed all other machine learning and deep learning models the authors built. The authors also sought to ensure the LIME explanations were correct, this was done by constructing a pipeline to find the Pointwise Mutual Information (PMI) of words throughout the entire dataset (in contrast to LIME which is local) and their association with a real or fake class. To conduct the evaluation, the 20 most important words for each post as found by LIME and extracted and compared to their rank in the PMI ranking using Kendall's Tau correlation coefficient. The Kendall's correlation for the BiLSTM was 0.35, the authors noted that the general range for Kendall's Tau coefficient was between 0.3 and 0.4, indicating that the LIME explanations had issues with alignment as the PMI context is global while LIME is local.

2.1.8 Fine-tuning Approaches

An issue with current models is that while they work well with data which was found in their training sets, they often suffer in performance when they encounter data unknown to them (Qin & Zhang, 2024). To solve this issue, Qin and Zhang (2024) proposed a novel approach, Feature Regularization-based Adversarial Fine-tuning (FGM-FRAT). This approach generates perturbed versions of the input in order to confuse the model. This forces the model to find new, deeper patterns instead of relying on otherwise superficial features. The key innovation however, is the addition of a regularization term to the loss function which seeks to minimise the difference between the features generated from the unperturbed, original text with the features generated from the perturbed text. This ensures the feature representation generated by the model will remain similar between original and perturbed text, improving generalisation. The authors used the FakeNewsNet dataset to fine-tune a BERT model using the FGM-FRAT approach using the FakeNewsNet dataset. The authors then tested the fine-tuned model with unseen datasets (BuzzFeedNews, LIAR, WELFake and KaggleFakeNews), using the standard BERT model as a baseline as well as some other fine-tuned BERT models which were fine-tuned with various approaches. The BERT-FGM-FRAT model performed the best amongst all models tested, with improvements of up to 0.32 in terms of F1-score against the base BERT model on the BuzzFeedNews dataset. While for other datasets the BERT-FGM-FRAT managed to offer significant performance improvements with the sole exception of the WELFake dataset where it was only able to improve upon the base model incrementally with a 0.03 (F1-score) improvement.

2.1.9 Graph-based Approaches

Methods which seek to analyse propagation structure for fake news detection often use graph-based methods which overlook the semantic information being propagated, while methods which analyse the semantic information of the content usually fail to understand the propagation structure (Z. Zhang et al., 2024). Research by Z. Zhang et al. (2024) proposed a novel approach, Graph Convolution Network and BERT combined with Co-Attention (GBCA) which seeks to combine graph based analysis of propagation techniques while using BERT for semantic analysis in order to classify fake news. To capture the spread of news, the proposed model constructs two graphs to capture propagation and dispersion using Edged-enhanced Bayesian Graph Convolution Network (EBGCN). For semantic feature extraction, the GBCA uses a pre-trained BERT to generate vectors from the content which capture the semantic data. And finally to integrate the two approaches, the authors used a co-attention mechanism which computes an affinity matrix to find the relationship between the features from both the structural (propagation and dispersion) and semantic features, the co-attention mechanism will then dynamically assign weights to the features. The authors used three datasets, Twitter15, Twitter16, and PHEME; and used baseline methods including GRU-RNN, SVM-TK (Support Vector Machine - Tree Kernel) among other graph-based methods and deep learning methods. GBCA managed to outperform all models it was compared against in terms of accuracy with a score of 92.8%, 90.3% and 70.8% for the Twitter15, Twitter16, and PHEME datasets respectively; however it was beaten by several Graph Convolution Networks (GCNs) when it came to F1-score for some of the classes. Despite this, it is worth noting the GBCA was more computationally efficient when training as the co-attention module used reduced the amount of iterations required for convergence.

2.2 Summary

Table 2.1: Summary of Literature Review Findings

Author(s)	Year	Title	Model Proposed	Proposed Methodology	Results	Notes	BERT used?
Q. Zhang et al. (2023)	2023	"A Deep Learning-based Fast Fake News Detection Model for Cyber-Physical Social Services"	Convolutional Feed-Forward (Conv-FFD)	Use of CNN to learn relationships and ultimately classify based on sentence-level embeddings generated by a "sliding window"	F1-scores of between 0.8702 - 0.8855.	Researchers one-hot-encoded Chinese characters, a technique not feasible for English text.	No
Saleh et al. (2021)	2021	"OPCNN-FAKE: Optimized Convolutional Neural Network for Fake News Detection"	Optimized Convolutional Neural Network to Detect Fake News (OPCNN-FAKE)	Use of GloVe to produce word embeddings which are then used to train the OPCNN-FAKE model.	F1-scores of between 0.959 - 0.9999	Extremely high scores on ISOT dataset indicate some level of overfitting.	No
Hashmi et al. (2024)	2024	"Advancing Fake News Detection: Hybrid Deep Learning With FastText and Explainable AI"	CNN-LSTM	Use of FastText to produce word embeddings which is then used to train a CNN-LSTM hybrid.	F1-scores of between 0.91 - 0.99	-	No
E.Almandouh et al. (2024)	2024	"Ensemble based High Performance Deep Learning models for fake news detection"	Bidirectional Long Short Term Memory – Bidirectional Gated Recurrent Unit (Bi-LSTM – Bi-GRU)	Researchers used FastText to generate embeddings which was used to train the hybrid model.	F1-scores of between 0.98 - 0.99	Research was done on Arabic text	No
Dhiman et al. (2024)	2024	"GBERT: A hybrid deep learning model based on GPT-BERT for fake news detection"	Generative Bidirectional Encoder Representations from Transformers (GBERT)	Use of BERT to generate word embeddings and GPT to generate global vectors. Outputs from BERT and GPT are subsequently combined and passed to a classifier.	F1-score of 0.9623	Researchers also experimented with XGBoost which performed nearly on par with GBERT in terms of accuracy.	Yes
Ali et al. (2022)	2022	"Deep Ensemble Fake News Detection Model Using Sequential Deep Learning Technique"	Deep Ensemble (Stacking)	Use of SDL models as base models and multilevel-perceptron classifier at the top.	F1-score of 1.00 for ISOT F1-scores of 0.48 - 0.82 for LIAR	LIAR dataset has more than 2 classes	No
Khanam et al. (2021)	2021	"Fake news detection using machine learning approaches"	Various Machine Learning Models	ML models fitted onto preprocessed datasets with key features extracted during processing	Accuracy of 0.73 (Best model; XGBoost)	-	No
Jing et al. (2023)	2023	"Multimodal fake news detection via progressive fusion networks"	Multimodal Progressive Fusion Approach (MPFN)	BERT for text feature extraction, visual feature extractor component for images and these two components are combined using the progressive multimodal feature fusion process.	F1-scores between 0.889 - 0.740	Multimodal approach	Yes
Xue et al. (2021)	2021	"Detecting fake news by exploring the consistency of multimodal data"	Multimodal Consistency Neural Network (MCNN)	A 5-module multimodal model using BERT for the text module, ResNet50 and ELA for the image modules with a final softmax classifier.	F1-scores between 0.968 - 0.831	Multimodal approach	Yes

Continued on next page

Table 2.1 (continued)

Author(s)	Year	Title	Model Proposed	Proposed Methodology	Results	Notes	BERT used?
Yuan et al. (2021)	2021	"Improving fake news detection with domain-adversarial and graph-attention neural network"	Domain-Adversarial & Graph Attention Neural Network (DAGA-NN)	Multimodal model using BERT and a Bi-LSTM for text and VGG19 with a neural network for images, features are then fused (features generated will not be domain-specific, due to adversarial training against the domain discriminator) and passed to the graph-attention-based-classifier.	F1-scores of 0.8828 and 0.9522	Multimodal with features trained adversarially to be domain-neutral	Yes
Wu et al. (2024)	2024	"Fake News in Sheep's Clothing: Robust Fake News Detection Against LLM-Empowered Style Attacks"	SheepDog	SheepDog leverages LLMs to remove stylistic differences in the dataset which is passed to a fine-tuned language model for classification.	0.7354 - 0.8563 (tested on perturbed dataset)	SheepDog is technically a framework	No
Szczepański et al. (2021)	2021	"New explainability method for Bert-based model in fake news detection"	DistilBERT-LIME (Unnamed by authors)	The proposed model used DistilBERT with several other layers for fake news classification, with Anchors and LIME being used to explain the otherwise black-box algorithms.	F1-score of 0.98	Used variant of BERT	Yes
Chien et al. (2022)	2022	"XFlag: Explainable Fake News Detection Model on Social Media"	XFlag	XFlag utilised Doc2Vec to extract features from the content, sentiment-features were extracted by a Google API, and user features from their profile. These were used to train an LSTM which was made explainable by the LRP algorithm.	F1-score of 0.938	Model considered more than just text and considered sentiment and user information as well.	No
M. Ahmed et al. (2022)	2022	"Explainable Text Classification Model for COVID-19 Fake News Detection."	Toxic Fake News Detection System (TFNDS)	Used Bi-LSTM in combination with LIME for explainability. Additionally attempted to evaluate accuracy of LIME using PMI.	F1-scores of 0.9425 and 0.894; Kendall's Tau Coefficient of 0.35	PMI evaluation likely had issues of alignment.	No
Qin and Zhang (2024)	2023	Boosting generalization of fine-tuning BERT for fake news detection	FGM-FRAT	Fine-tuning using an adversarial approach can substantially improve generalisation of models	Improvements in performance of between 0.32 and 0.03 (F1-score) relative to base BERT model	-	No
Z. Zhang et al. (2024)	2024	"GBCA: Graph Convolution Network and BERT combined with Co-Attention for fake news detection"	Graph Convolution Network and BERT combined with Co-Attention (GBCA)	GBCA builds graphs to capture the dissemination of news and BERT to process the content. These approaches are fused together with a co-attention mechanism.	Accuracy of 92.8% - 70.8%	-	Yes

3 Theoretical Framework

3.1 Deep Learning Approaches

3.1.1 Convolutional Neural Networks

Convolutional Neural Networks or CNNs are neural networks which generally have 3 layers: A convolutional layer, pooling layer and a fully-connected layer. The convolutional layer is the key part of a CNN, and it uses a grid of numbers called kernels (or filters) to 'scan' through the various features and perform a mathematical operation known as 'convolution'. After the convolution layer, the pooling layer reduces the size of the data which also helps with reducing the computational demands and also helps with ensuring the model doesn't become focussed on small and irrelevant details. The final, fully-connected layer works by flattening the prior layer's multi-dimensional features into a single layer. This single-layer output can now be fed to a standard neural network for classification.

3.1.2 Long Short Term Memory

Long Short Term Memory of LSTMs is a type of Recurrent Neural Network (RNN) which aims to solve an issue with RNNs called the 'Vanishing Gradient Problem'. Essentially, the issue is that RNNs work using gradients which inform the weights in neurons, and these gradients are calculated using the chain rule, which means after dozens of multiplications a gradient will rapidly lose its value. This means that RNNs have 'short term memory', an issue LSTM's solve by selectively holding on to important information in a 'Cell State' through a system of gates. There are three gates: forget gate, input gate, and output gate. The forget gate figures out what information should be forgotten because it is now longer useful. The input gate decides what is worth adding to the 'Cell State'. And lastly the output gate figures out what information in the 'Cell State' is important or relevant to the current situation and outputs that. This ensures that LSTMs are able to hold on to important information over a long-term and doesn't suffer from short-term forgetfulness. This makes LSTMs especially suitable for sequential input where important discriminating features are far between each other.

3.2 Embedding Approaches

3.2.1 Word2Vec

Word2Vec is a neural network embedding model which converts words into dense vector representations developed by Mikolov et al. (2013). There are two main architectural variants, Continuous Bag of Words and Skip-gram; Skip-gram was used in this instance and thus will be the only variant elaborated on.

The Word2Vec Skip-gram variant works by trying to predict the words around a single target word. So, using the example of the sentence "I love vanilla ice cream", the target word being the word "vanilla" and the windows being set at two words; the model would focus on trying to predict the words "I", "love", "ice", "cream" around the word "vanilla".

To do this, the input word (represented in on-hot-encoded form) is taken in by the model as input and passed to the hidden layer and the outputs from this layer is passed to the output layer which tries to guess the words surrounding the input. If the model gets it right, say by correctly guessing the words "ice" and "cream", the connections between these words and the target word is strengthened and if it fails the model will update the weights in the hidden layer so that it will be more likely to guess the correct words in the next iteration.

This massive guessing operation is used to train the hidden layer, so that when the training is complete, the output layer is stripped and the output of the hidden layer becomes the vectors. So to conclude, Word2Vec's Skip-gram variant works by trying to guess words around a target word and through this exercise trains a hidden layer which will be used to turn words into vectors.

3.2.2 Bidirectional Encoder Representations from Transformers

Bidirectional Encoder Representations from Transformers or BERT is a self-supervised bidirectional model which differs from Word2Vec and GloVe in that while Word2Vec have a single vector for each word; BERT outputs different vectors for each word depending on the context.

To do this, BERT is trained through two objectives: Masked Language Modelling (MLM) and Next Sentence Prediction (NSP) (Liu et al., 2019). MLM works by trying to get BERT to predict the missing word. So using the previous example of "I love vanilla ice cream", say the word "ice" is 'masked' (omitted), BERT will have to try and guess the missing word from: "I love vanilla [mask] cream". Since BERT is bidirectional, it will consider words preceding and subsequent to the mask. This technique gets BERT to learn the relationship between words in a sentence.

The next technique, NSP is exactly as it sounds. It tries to get BERT to guess if a sentence comes directly after another sentence (Liu et al., 2019). So, let's say BERT is given two sentences: the previous example of "I love vanilla ice cream.", and "It tastes so sweet". BERT will have to guess if the latter sentence comes after the former. This was intended to get BERT to learn relationships between pairs of sentences.

However it is important to note that the importance of NSP to BERT may be overstated, as Liu et al. (2019) hypothesised after experiments with the NSP component of the loss function showed that removing the NSP loss function could match or even slightly improve performance on downstream tasks.

As for how BERT is employed for classification, a linear classifier is employed. This linear classifier takes the CLS token which is a 768 dimension representation of the input and outputs prediction scores which are passed through a softmax function to get the final classification. During fine-tuning, the difference between predicted probabilities and true labels is used to calculate the cross-entropy loss and the linear classifier as well as the BERT model weights are updated via backpropagation.

In conclusion, the bidirectional nature of BERT gives it an edge and the objectives used to train it, specifically MLM gave it an advantage when understanding context, allowing it to create more meaningful vectors. As a note, BERT does have many variants, including RoBERTa, developed by Liu et al. (2019) who were cited earlier; this research uses the original BERT, developed by Google.

3.3 Explainable AI Approaches

3.3.1 LIME

Local Interpretable Model-Agnostic Explanations or LIME is a technique used to explain black-box models. LIME, true to its acronym tries to explain the black-box model 'locally', by taking a prediction, then creating many variations of the data point which lead to that prediction to see how the black-box model responds to these changes. These 'perturbed' data points are assigned weights, with the greater in similarity a point is to the original data point, the higher the weight. These perturbed data points and the black-box's prediction are then used to form a small, weighted dataset which is fitted onto a small interpretable model (often linear regression). The weights from this model will be used to explain the behaviour of the actual black-box model for that one specific prediction.

To put the above into an example, let's say there is a XGBoost model being used to determine if a person has diabetes. Let's also say that this model only takes 3 features as input: weight, height and age. When given the input [90, 169, 56], it predicts true. To explain this, LIME will then generate variations around this input data, so called 'perturbed data points', for example [89, 167, 58]. These variations of the original data points are then passed to the XGBoost model as input and the output is noted. A dataset is then constructed based on the perturbed data points, the black-box's output, and the proximity of the perturbed data points to the original data point. This weighted dataset is fit onto a linear regression model and this model, as it is interpretable and not a black-box, is used to explain the black-box's prediction.

4 Research Methodology

4.1 Datasets

4.1.1 WELFake Dataset

The WELFake dataset was downloaded from Kaggle, and contains 72134 rows of data. The class balance was relatively balanced, with the real news class having an excess of 2000 rows as can be referred to in Figure **??**. As for features, the dataset came with four columns, an unnamed ID column, title (of the news article), text (from the news article) and the label (real/fake). There were 558 missing values in the 'title' column and 39 missing values in the 'text' column initially.

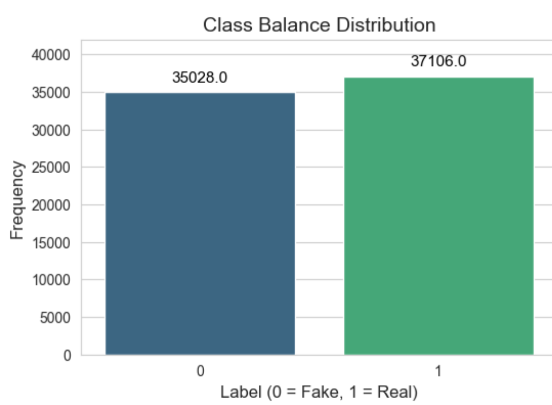

Figure 4.1: Class Distribution of the WELFake Dataset

4.1.2 Fake News Classification

This dataset, named "Fake News Classification" was downloaded from Kaggle. It contains 40597 rows of data with 4 columns: an unnamed ID column, title, text and label. There were no missing values or duplicates in any of the rows initially before pre-processing. And the class balance had an excess of approximately 3000 in favour of the positive class (real).

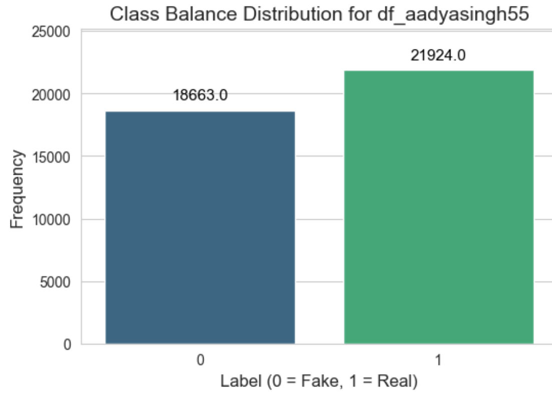

Figure 4.2: Class Distribution of the Fake News Classification Dataset

4.1.3 FakeNewsNet

The FakeNewsNet dataset was downloaded from Kaggle, it is worth noting this was not the original dataset, it had its various Comma Separated Values (CSV) files merged by the uploading user "algord". The original dataset was gathered and used for research by Shu et al. (2018), Shu, Wang, and Liu (2017) and Shu, Sliva, et al. (2017). We thank the authors for developing and providing this dataset for use. FakeNewsNet contains 23196 rows, with 5 columns: title, news_url, source_domain, tweet_num and label. The dataset contained missing values in the news_url and source_domain columns (which is acceptable as this is not required later on in training or testing). There were 137 duplicate rows (which will be dropped later on) initially. The class balance was heavily skewed in favour of the positive (real) class, with approximately 75% of all rows belonging to the positive class.

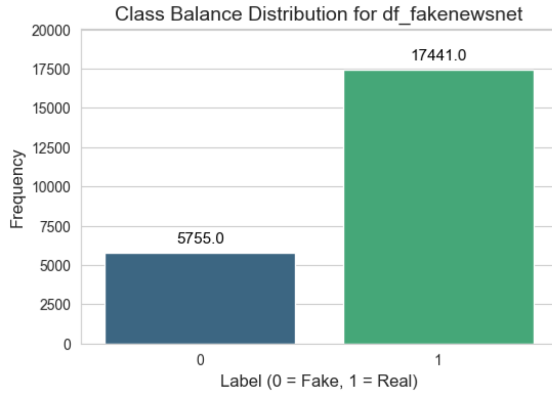

Figure 4.3: Class Distribution of the FakeNewsNet Dataset

4.1.4 Fake News Detection

The Fake News Detection dataset was downloaded from Hugging Face. It contains 44267 rows and 6 columns: Unnamed ID column, title, text, subject, date, and label. There were no missing values or duplicate rows initially. The class balance is relatively balanced with the difference between the positive and negative classes being approximately ± 3%.

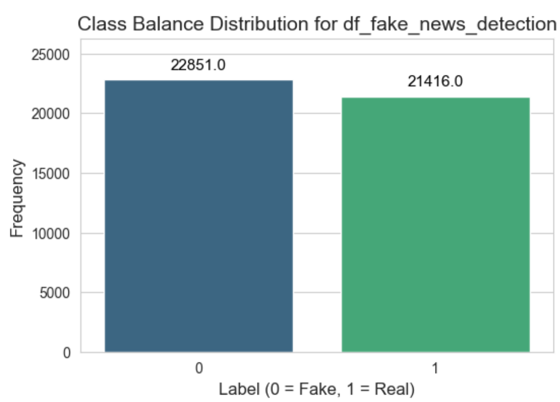

Figure 4.4: Class Distribution of the Fake News Detection Dataset

4.1.5 ISOT

The ISOT dataset was downloaded from the University of Victoria website. It was originally gathered and used for research by H. Ahmed et al. (2018) and H. Ahmed et al. (2017). We thank the authors for gathering this dataset and providing it. It contains 44898 rows with 5 columns: title, text, subject, date and label. There were no missing values initially. However, there were 209 duplicate rows initially. The class balance was relatively balanced, with the difference in proportion of the positive and negative classes being approximately ± 5%.

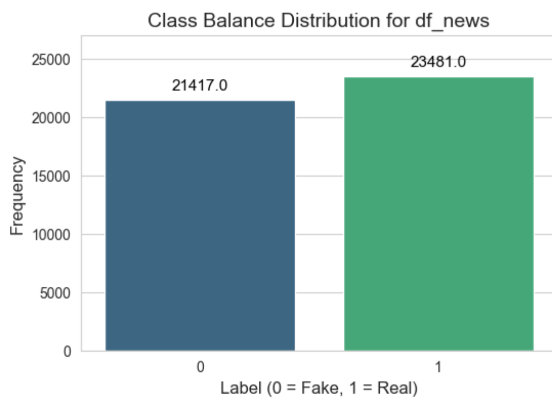

Figure 4.5: Class Distribution of the ISOT Dataset

4.2 Data Understanding

4.3 Data Pre-processing

The data was processed by first merging 'title' and 'text' columns together if the dataset had split 'title' and 'text' into different columns, this would be combined. In one of the datasets "FakeNewsNet", only the 'title' column was used. After the columns were combined, the new 'combined_text' column was checked for missing strings and null values, of which none were found.

After combining the columns, any columns that did not contain the label or 'combined_text' column were dropped, such as source URLs columns, and other metadata like date. This revealed substantially more duplicated rows, likely as a result of the datasets being gathered through scraping methods which scraped the same content from multiple sources causing the dataset to have duplicated textual columns but different metadata, which allowed it to evade detection earlier. The total number of samples dropped is summarised in Table **??**.

Once completed, the 'combined_text' column was stripped of any non-Unicode characters as well as links, phone numbers and other non-ASCII characters. The removal of non-ASCII characters is possible as this research focuses on English news. After this, any rows with empty strings are removed.

Table 4.1: Summary of Data Preprocessing Statistics

	Initial		After Combining Text		After Dropping Rows		After Cleaning	
Dataset	**Missing***	**Duplicates**	**Missing***	**Duplicates**	**Missing***	**Duplicates**	**Missing***	**Duplicates**
WELFake	597	0	0	-	-	8458	6	-
Fake News Classification	0	0	0	-	-	2	5	-
FakeNewsNet	660	137	0	-	-	1349	3	-
Fake News Detection	0	0	0	-	-	5612	5	-
ISOT	0	0	0	-	-	5795	5	-

Note. (-) indicates metric not checked at this stage.
() Shows the sum of missing values, not necessarily the number of rows (in stages after Initial, only `combined_text` is checked).

Table 4.2: Dataset Size Changes After Preprocessing

	Initial		Final	
Dataset	**Rows**	**Columns**	**Rows**	**Columns**
WELFake	72,134	4	63,670	2
Fake News Classification	40,587	4	40,580	2
FakeNewsNet	23,196	5	21,844	2
Fake News Detection	44,267	6	38,650	2
ISOT	44,898	5	39,098	2

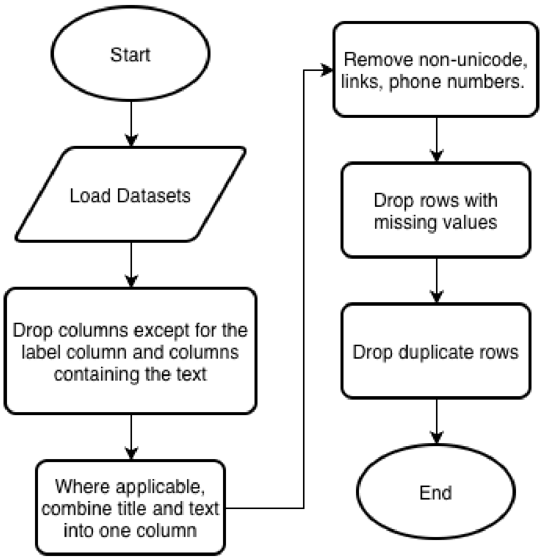

Figure 4.6: Data Preprocessing Steps Flowchart

After the above processes, the data was then processed separately depending on the type of model being trained. For BERT, the data was tokenised using BERT's tokeniser without further processing steps. For Word2Vec, the data was both decapitalised and lemmatised before the data is used to train the Word2Vec model.

Table 4.3: Summary of Additional Preprocessing Steps for Each Model Type

Model Type	**Preprocessing Steps**
BERT	Tokenisation using BERT's tokeniser (no additional preprocessing)
Word2Vec	• Decapitalisation (lower-case conversion) • Lemmatisation • Splitting into individual words

4.4 Experimental Setup

Each dataset, for all models (BERT, TPA-BERT, ML and DL models), was split 85-15, 85% training and 15% test. The 85% training set is further split 80-20, train and validation respectively, using 5-fold cross-validation. The 15% test set is used at the end of the training process for final evaluation of each fold.

As this research focuses on generalisation, another evaluation step is conducted: "cross-dataset evaluation". Cross-dataset evaluation is the term we will use to refer to the process where a model is tested against completely unseen datasets, for example: a model trained on WELFake is tested against FakeNewsNet and ISOT. Only one fold is selected for cross-dataset evaluation, specifically the fold which performed best based on validation F1-score.

Below is a summary of the modelling approaches for each embedding type (BERT and Word2Vec). BERT is fine-tuned, while Word2Vec has its embeddings fit onto ML and DL models.

Table 4.4: Summary of Modelling Approaches for Each Embedding Type

Embedding Type	**Modelling Approach**
BERT	• Fine-tuning to convert BERT into a classification model
Word2Vec	• Machine learning models (trained on vectorised datasets) • Deep learning models: CNN, LSTM, CNN-LSTM

4.4.1 BERT Fine-tuning

Experiments using BERT involved fine-tuning a base BERT-uncased model. The datasets were previously cleaned and processed as was explain previously, but in order for the data to be suitable to input to BERT, it undergoes further processing in the form of tokenisation using the base BERT-uncased tokeniser with the settings shown in Table **??**.

Table 4.5: BERT Tokeniser Configuration

Parameter	**Value**
Truncation	True
Padding	True
Max Length	128

The reason for capping the `max_length` at 128 instead of at 512 (max) were due to several factors. Firstly it was hoped that by limiting the maximum length it would reduce the amount of overfitting, improving generalisation. And secondly a lower `max_length` resulted in lower computational demands, especially on memory.

After tokenisation, the BERT-uncased model is loaded and the training arguments are set. The training arguments are shown in Table **??**.

Table 4.6: Baseline BERT Fine-tuning Configuration

Parameter	**Value**
Number of Epochs	3
Train Batch Size (per device)	128
Eval Batch Size (per device)	128
Eval Strategy	Steps
Eval Steps	128
Save Strategy	Steps
Save Steps	128
Save Total Limit	2
Optimizer	AdamW (PyTorch)
Logging Steps	128
Metric for Best Model	F1
Load Best Model at End	True
Weight Decay	0.01

The number of epochs was set at 3, to avoid overfitting but just enough to allow for the weights to be updated to perform better and the batch size was set to 128 to allow for training to progress at a reasonable rate while also allowing for stable gradient updates. The high batch rate was primarily chosen due to limited computational resources, regardless as will be shown later the results remain quite high despite the high batch rate.

At the end the best model is selected using validation F1-score, and this model will undergo cross-dataset evaluation to determine its generalisation performance.

4.4.2 Word2Vec Deep Learning and Machine Learning

Word2Vec experimentation saw the text (from previous processing steps) undergo some final processing which involved turning the text into all lower-case and the text undergoing lemmatisation using the Python library NLTK's *WordNetLemmatizer*. After this, the processed text was used to train the Word2Vec model for each dataset (as there are 5 datasets, there will be 5 different Word2Vec embedding models).

The training arguments for the Word2Vec models can be found in **??**.

Table 4.7: Word2Vec Models' Training Configuration

Parameter	**Value**
Vector Size	300
Window	5
Min Count	2
Workers	1
Epochs	10
Training Algorithm	Skip-gram

After each Word2Vec model is trained, the models are employed to convert each dataset into embedding form which can be understood by machine learning and deep learning models. After this process, each respective dataset and its embeddings were then used to train several machine learning models (KNN, Gaussian NB, Logistic Regression, SVC, XGBoost, Random Forest) and deep learning models (CNN, CNN-LSTM, LSTM).

Below are tables containing the training parameters for the models trained (note: a seed of 42 was used for reproducibility):

Table 4.8: Machine Learning Model Training Hyperparameters

Model	**Training Arguments**
K-Nearest Neighbors (KNN)	`n_neighbors=5`
Gaussian Naive Bayes	Default parameters
Logistic Regression	`max_iter=2000, class_weight='balanced', random_state`
Support Vector Classifier (SVC)	probability=True, class_weight='balanced', random_state=se
XGBoost	`eval_metric='logloss', random_state=seed, n_estimato`
Random Forest Classifier	`n_estimators=300, class_weight='balanced', random_st`

Table 4.9: Deep Learning Model Hyperparameters

Model	**Parameter**	**Value**
	Common Training Settings	
	Epochs	10
	Batch Size	32
	Learning Rate	0.001
	Optimizer	Adam
	Loss Function	BCEWithLogitsLoss
	Early Stopping Patience	3
	Max Sequence Length	200
	Embedding Layer	Pre-trained Word2Vec (300-d)
TextCNN	Number of Filters	128
	Filter Sizes	[3, 4, 5]
	Dropout	0.5
TextLSTM	Hidden Dimension	128
	Number of Layers	2
	Dropout	0.5
TextCNN-LSTM	CNN Filters	128
	CNN Kernel Size	3
	LSTM Hidden Dims	128
	LSTM Layers	1
	Dropout	0.5

After each fold completes training, it is evaluated against the 15% test set. Then after all the folds have completed training, the best fold according to validation f1-score is chosen for cross-dataset evaluation.

4.4.3 Proposed Model

The proposed model, Text Perturbation-based Adversarial BERT (TPA-BERT), sought to improve generalisation when it came to fake news detection, particularly against unseen data. This was to be done by directly perturbing the text, followed by adversarial training to minimise the difference in embeddings generated when given original and perturbed data. The intention is, by scrambling the text, and with the adversarial training minimising the differences in model output, TPA-BERT would generate embeddings which are more robust and learn to identify more 'in-depth' features for fake news detection.

Going in more depth regarding the methodology: firstly, before any training began, the datasets are perturbed using NLTK's WordNet by loading each record and applying synonym substitution against 20% of each record's words, followed by another 10% of words which are randomly swapped (swapped between words in the same row) and lastly 10% of words are randomly deleted. Secondly, the BERT model is loaded, and both the original and perturbed datasets undergo the dataset splitting applied to all models (described previously). The data is then tokenised according to the settings described in **??**. Following this, the training begins with the model accepting input and beginning to learn from the adversarial loss function (**??**) which combines the cross entropy loss and cosine similarity loss (the difference between the CLS tokens produced by the model for the original and perturbed data input). Note that the cosine similarity loss is linearly scaled from 0 to 0.5 over the first 10% of training steps. Once training is complete, whether by completing all epochs, or by early-stopping (set at 3 steps) the model is evaluated on the held out test set. And to test the model's generalisation, the best model according to validation f1-score is chosen for cross-dataset evaluation.

TPA-BERT loss function:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{adv}} \times \mathcal{L}_{\text{cosine}} \tag{1}$$

Table 4.10: TPA-BERT Fine-tuning Configuration

Parameter	**Value**
Number of Epochs	3
Train Batch Size (per device)	32
Eval Batch Size (per device)	32
Gradient Accumulation Steps	2
Eval Strategy*	Steps
Eval Steps*	128
Save Strategy	Steps
Save Steps	128
Save Total Limit	1
Optimizer	AdamW (PyTorch)
Logging Steps	128
Metric for Best Model	F1
Load Best Model at End	True
Weight Decay	0.01

*Although Eval Steps and Eval Strategy are not explicitly set in the code, by default the strategy is “steps” and the steps are “128”.

There are several modifications to the parameters compared to the baseline BERT model. Most notably the training batch size which is now 32 instead of the 128 used by the baseline. There are a few reasons for this: firstly, the model also has to process the adversarial data which essentially doubles the data processed. And secondly, because of the aforementioned doubled data, the forward and backward passes now carry twice as much data. As a result, with double the data as well as the fact the forward and backward passes also have to deal with double the data, the effective batch size is 128.

The proposed model is inspired by and similar to the process described by Qin and Zhang (2024), however, unlike the model described by those authors, TPA-BERT perturbs the text directly, instead of at the embedding and feature level like in FGM-FRAT (the model described by the authors). The reason for this deviation is that perturbation at a text level is simple to implement, in addition to the decreased number of passes per step thus reducing computational demands. Furthermore, the proposed model also contains a LIME component which will complement the BERT component by giving explanations for the otherwise black box model to increase trust in the model.

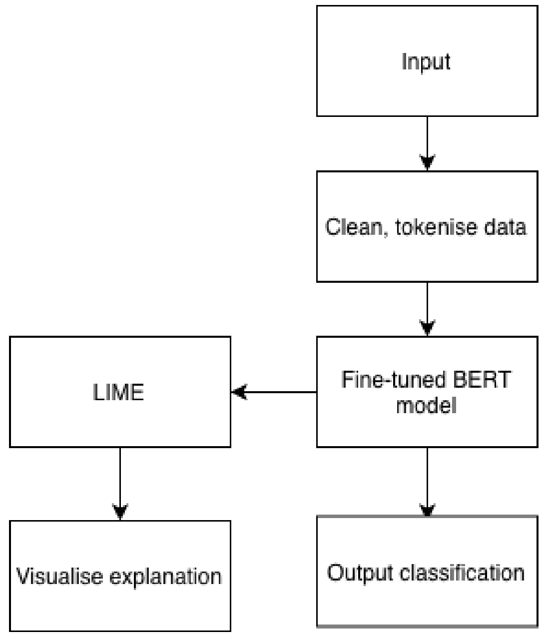

Figure 4.7: Proposed TPA-BERT Model Architecture

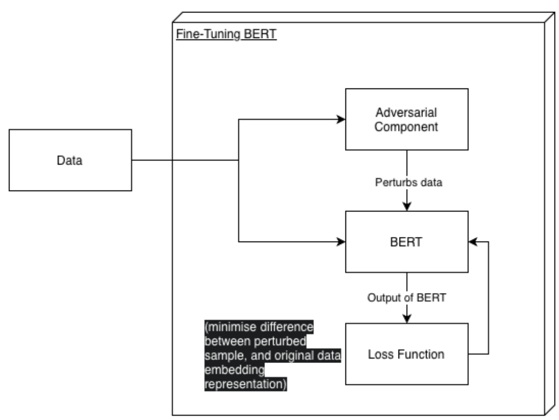

Figure 4.8: Proposed Fine-Tuning Process for TPA-BERT

4.4.4 LIME

LIME was implemented on BERT and TPA-BERT. For this implementation, LIME generates a local dataset of 1000 perturbed samples of the input by randomly removing words from the original input. This local dataset is then passed through the classification model (BERT or TPA-BERT) to obtain the prediction probabilities.

As the BERT tokeniser is capped at 128 tokens, to ensure consistent output the perturbed data is tokenised with static padding at 128 tokens. Esnuring a fixed size of 128.

Once the probabilities were obtained, LIME fits a linear model onto the weighted perturbations. Weights are determined by cosine distance to the original text. For this study, the top 20 features were selected to explain the model's behaviour.

4.5 Evaluation

Evaluation of the models will be done by comparing F1-score, and its constituent metrics of recall and precision, as well as accuracy. These metrics will be obtained from the trained models by using the test sets.

F1-score is a formula which combines precision and recall. Precision describes the proportion of a model's positive predictions and is calculated by dividing the number of *True Positives* by the sum of *True Positives* and *False Positives*. Recall describes the model's ability to identify positive instances and is calculated by dividing the number of *True Positives* by the sum of *True Positives* and *False Negatives*. Together these metrics are combined to form F1-score, which is calculated by dividing *2* by the sum of the inverse of *Precision* and *Recall*.

Meanwhile, accuracy is found by dividing the number of correct predictions over the total number of predictions.

The formulae described above are also written below in mathematical notation:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{2}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{3}$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}} \tag{5}$$

4.5.1 LIME Evaluation

To evaluate the implementation of LIME, two metrics were used: Average Attribution Strength and "Faithfulness". Average Attribution Strength refers to the average (absolute) LIME attribution scores for the top 20 features. And Faithfulness is calculated by removing the top K (evaluated with K at 5, 10 and 20) features identified by LIME and then finding the difference in the confidence of the prediction between the original input and the top features-removed input.

The specific formulae are below:

$$\text{Attribution Strength} = \frac{1}{N}\sum_{i=1}^{N} |w_i| \tag{6}$$

where $N = 20$ represents the number of top features, and w_i denotes the LIME attribution coefficient assigned to feature i.

$$\text{Faithfulness}(K) = P(y \mid X_{\text{orig}}) - P(y \mid X_{\text{mod}}^{(K)}) \tag{7}$$

where:

- y represents the model's predicted class for the original input text X_{orig}.
- $P(y \mid X)$ is the model's predicted probability (confidence) for class y given the input X.
- $X_{\text{mod}}^{(K)}$ is the modified input text after deleting the top K words (evaluated at $K \in \{5, 10, 20\}$) with the largest absolute attribution weights $|w_i|$.

5 Results

5.1 BERT

Table 5.1: Baseline BERT Results

Dataset Name	Accuracy	Precision	Recall	F1
WELFake	0.9917	0.9920	0.9897	0.9908
FakeNewsNet	0.8497	0.8768	0.9325	0.9037
Fake News Detection	0.9976	0.9986	0.9971	0.9978
ISOT	0.9996	0.9991	1.0000	0.9995
Fake News Classification	0.9934	0.9948	0.9929	0.9938

The results from the baseline fine-tuned BERT models exhibited very high accuracy, recall and precision. As can be referred to in Table **??**, it appears that the FakeNewsNet dataset was the most challenging dataset. All the other datasets proved to be no challenge for each of their fine-tuned BERT models.

Table 5.2: Cross Dataset Generalisation (BERT) - WELFake Only

Dataset Name	Accuracy	Precision	Recall	F1
FakeNewsNet	0.6984	0.7562	0.8872	0.8165
Fake News Detection	0.2231	0.3302	0.4054	0.3640
ISOT	0.9992	0.9998	0.9983	0.9991
Fake News Classification	0.0188	0.0380	0.0336	0.0357

The best fold by validation f1-score was selected for cross dataset evaluation. The results show that the model trained on WELFake performed quite well for FakeNewsNet and ISOT, and much poorer for Fake News Detection and Fake News Classification datasets.

5.2 Deep Learning

Table 5.3: Deep Learning Baseline Results

Dataset Name	Model	Accuracy	Precision	Recall	F1
WELFake	CNN	0.9610	0.9504	0.9645	0.9573
	LSTM	0.9489	0.9354	0.9536	0.9442
	CNN-LSTM	0.9474	0.9433	0.9408	0.9419
FakeNewsNet	CNN	0.8395	0.8689	0.9280	0.8974
	LSTM	0.7565	0.7565	1.0000	0.8614
	CNN-LSTM	0.7565	0.7565	1.0000	0.8614
Fake News Detection	CNN	0.9883	0.9881	0.9907	0.9894
	LSTM	0.9824	0.9806	0.9876	0.9841
	CNN-LSTM	0.9832	0.9839	0.9855	0.9847
ISOT	CNN	0.9983	0.9976	0.9986	0.9981
	LSTM	0.9959	0.9942	0.9969	0.9955
	CNN-LSTM	0.9979	0.9975	0.9980	0.9977
Fake News Classification	CNN	0.9901	0.9959	0.9858	0.9908
	LSTM	0.9852	0.9910	0.9816	0.9862
	CNN-LSTM	0.9862	0.9922	0.9822	0.9872

The results for the deep learning models trained using the Word2Vec embeddings indicate that all the models had similar performances and were generally on par with one another, although CNNs very slightly outperformed the LSTMs and CNN-LSTMs. All models achieved results exceeding an f1 score of 0.94, with the exception of models trained on FakeNewsNet.

Table 5.4: Cross Dataset Generalisation (CNN and LSTM Only) - Trained on WELFake Only

Dataset Name	Model	Accuracy	Precision	Recall	F1
FakeNewsNet	CNN	0.6932	0.6274	0.6932	0.6516
	LSTM	0.7063	0.6329	0.7063	0.6573
Fake News Detection	CNN	0.1514	0.1403	0.1514	0.1456
	LSTM	0.0269	0.0288	0.0269	0.0276
ISOT	CNN	0.9959	0.9959	0.9959	0.9959
	LSTM	0.9901	0.9902	0.9901	0.9901
Fake News Classification	CNN	0.0222	0.0242	0.0222	0.0231
	LSTM	0.0278	0.0292	0.0278	0.0283

*due to the results from CNN and LSTM models generally outperforming the CNN-LSTM hybrid, their results have been omitted for brevity.

Looking at the cross-dataset evaluation results, a similar pattern to the BERT's cross-dataset evaluation emerges. With performance for ISOT and FakeNewsNet being excellent when compared to performance for Fake News Detection and Fake News Classification.

5.3 Machine Learning

Table 5.5: Machine Learning Baseline Results

Dataset Name	Model	Accuracy	Precision	Recall	F1
WELFake	SVC	0.9501	0.9412	0.9492	0.9452
	Logistic Regression	0.9319	0.9219	0.9286	0.9252
	XGBoost	0.9304	0.9288	0.9169	0.9228
	Random Forest	0.8961	0.8930	0.8758	0.8843
	KNN	0.8725	0.9234	0.7839	0.8480

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Table 5.5 (continued)

Dataset Name	**Model**	**Accuracy**	**Precision**	**Recall**	**F1**
	Gaussian NB	0.6998	0.7932	0.4572	0.5800
FakeNewsNet	SVC	0.8057	0.8965	0.8401	0.8674
	Logistic Regression	0.7841	0.9031	0.8005	0.8487
	XGBoost	0.8255	0.8437	0.9442	0.8911
	Random Forest	0.8226	0.8251	0.9714	0.8923
	KNN	0.8097	0.8382	0.9275	0.8806
	Gaussian NB	0.7643	0.8535	0.8311	0.8421
Fake News Detection	SVC	0.9861	0.9830	0.9919	0.9874
	Logistic Regression	0.9830	0.9825	0.9867	0.9846
	XGBoost	0.9752	0.9698	0.9855	0.9776
	Random Forest	0.9529	0.9422	0.9738	0.9577
	KNN	0.9456	0.9330	0.9704	0.9514
	Gaussian NB	0.8971	0.9122	0.8989	0.9055
ISOT	SVC	0.9878	0.9903	0.9830	0.9867
	Logistic Regression	0.9869	0.9877	0.9838	0.9857
	XGBoost	0.9753	0.9811	0.9647	0.9728
	Random Forest	0.9559	0.9667	0.9359	0.9511
	KNN	0.9479	0.9708	0.9137	0.9414
	Gaussian NB	0.9093	0.8935	0.9104	0.9019
Fake News Classification	SVC	0.9759	0.9824	0.9727	0.9775

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Table 5.5 (continued)

Dataset Name	Model	Accuracy	Precision	Recall	F1
	Logistic Regression	0.9701	0.9733	0.9713	0.9723
	XGBoost	0.9638	0.9651	0.9679	0.9665
	Random Forest	0.9449	0.9450	0.9535	0.9492
	KNN	0.9336	0.9249	0.9546	0.9395
	Gaussian NB	0.8989	0.9195	0.8909	0.9050

The machine learning models generally did very well on all datasets save for FakeNewsNet. When compared against BERT, BERT outperformed the ML results at a degree similar to how it outperformed the DL results; and when compared against DL results, the DL models (or CNN, which was the best DL model) only very slightly outperformed the best ML models for each dataset.

Table 5.6: Cross Dataset Generalisation (SVC, Random Forest and XGBoost ONLY) - WELFake Only

Dataset Name	Model	Accuracy	Precision	Recall	F1
FakeNewsNet	SVC	0.7536	0.6022	0.7536	0.6509
	Random Forest	0.6855	0.6291	0.6855	0.6508
	XGBoost	0.6963	0.6169	0.6963	0.6459
Fake News Detection	SVC	0.0192	0.0181	0.0192	0.0184
	Random Forest	0.0190	0.0166	0.0190	0.0176
	XGBoost	0.0183	0.0166	0.0183	0.0172
ISOT	SVC	0.9841	0.9843	0.9841	0.9841
	Random Forest	0.9822	0.9825	0.9822	0.9821
	XGBoost	0.9844	0.9847	0.9844	0.9844
Fake News Classification	SVC	0.0333	0.0342	0.0333	0.0335

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Table 5.6 (continued)

Dataset Name	Model	Accuracy	Precision	Recall	F1
	Random Forest	0.0345	0.0347	0.0345	0.0343
	XGBoost	0.0332	0.0336	0.0332	0.0332

*Models were selected due to generally outperforming the other models and for brevity's sake.

The cross-dataset evaluation of the selected ML models again followed the pattern observed by the BERT and DL model's cross-dataset evaluation results. However, most notably for the ML models the results for Fake News Detection was much poorer than BERT and the DL models. For reference, BERT and CNN achieved an f1 score of 0.3640 and 0.1456 respectively, while the best ML model (for Fake News Detection): SVC, only achieved an F1-score of 0.0184.

As for the other datasets, the ML models were very slightly outperformed by the DL models, with such a small difference than randomness could have very well caused it. With the exception of Fake News Classification, but the results for that dataset are already quite poor and as a result is of little consequence.

5.4 TPA-BERT

Table 5.7: TPA-BERT Results

Dataset Name	Accuracy	Precision	Recall	F1
WELFake	0.9882	0.9852	0.9888	0.9870
FakeNewsNet	0.8425	0.8702	0.9305	0.8994
Fake News Detection	0.9947	0.9933	0.9971	0.9952
ISOT	0.9993	0.9987	0.9998	0.9993
Fake News Classification	0.9890	0.9938	0.9858	0.9898

The results for TPA-BERT were excellent, with results exceeding F1-scores of 0.95 with the exception of FakeNewsNet. When comparing the results against BERT, BERT again very slightly (<0.005 difference in f1 score) outperforms TPA-BERT. Overall these results is very typical when observed together with the other results. The adversarial training does seem to have very slightly impacted the performance of the model but not by much.

Table 5.8: Cross Dataset Generalisation (TPA-BERT) - WELFake Only

Dataset Name	Accuracy	Precision	Recall	F1
FakeNewsNet	0.6996	0.7520	0.8995	0.8191
Fake News Detection	0.2178	0.3246	0.3946	0.3562
ISOT	0.9982	0.9994	0.9968	0.9981
Fake News Classification	0.0196	0.0383	0.0338	0.0359

The cross-dataset evaluation for TPA-BERT followed the usual pattern described earlier. The results however, when compared against BERT were only slightly better (<0.008 difference in f1 score). Even then this small difference may be attributed to randomness. Regardless, these results indicate that the adversarial technique used in TPA-BERT failed to generalise the model sigificantly and is on par with the BERT results.

5.5 LIME Results

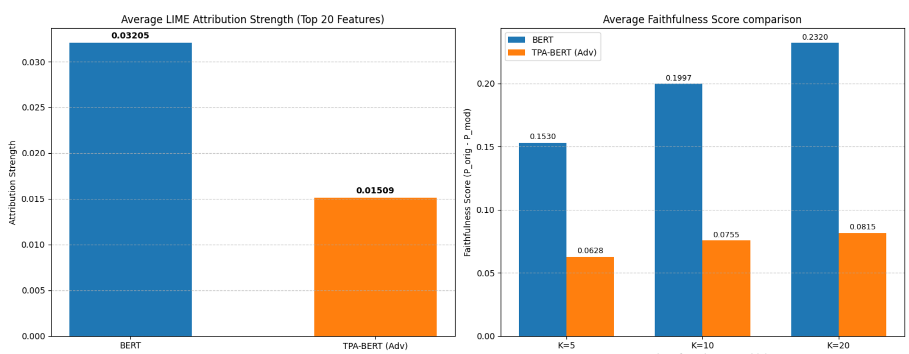


Figure 5.1: LIME Evaluation Results

LIME was implemented on the BERT and TPA-BERT models trained on WELFake. And the evaluation was done on these two models.

The results indicated that the average attribution scores were stronger for BERT than TPA-BERT, with average scores of 0.032 and 0.015 respectively. This indicates that LIME was better able to explain the BERT model than the TPA-BERT model.

As for the faithfulness score, it was evaluated with K = 5, 10 and 20. The results indicate some level of correlation between the number of features removed and the faithfulness score. Furthermore, the results indicate that the features LIME identified do in fact meaningfully impact the classification. When comparing BERT and TPA-BERT, it quickly becomes evident that TPA-BERT is less faithful than BERT.

6 Discussion

6.1 Model Performance Comparison

Comparing the performance of all the models trained (BERT, DL and ML models), BERT outperforms all other models, if only slightly in some cases. Although it is worth mentioning this difference ranges from small (F1 score improvement of 0.03 for CNN trained on WELFake) to insigificant. Furthermore, BERT does not appear to have fallen into the trap of classifying only positives for FakeNewsNet like the LSTM and CNN-LSTM models fell into. Although the F1-score improvement is negligible (<0.01), it does show that BERT is able to produce embeddings which are more descriptive than Word2Vec.

The DL results for FakeNewsNet (as mentioned above) are rather interesting, as by looking at the recall we can see that the LSTM and CNN-LSTM models achieved perfect recall. This is most likely because there is a significant class imbalance in favour of the positive class. The LSTM and CNN-LSTM models seem to have fallen into the trap of always classifying any input as positive, which due to the class imbalance does mean that it actually performs quite well. The CNN model seems to have escaped this, as it does not have a perfect recall but has better precision, indicating it is actually classifying the negative class with some level of accuracy as well.

Like the DL and BERT models, the performance of the ML models were exceptional, with the exception of FakeNewsNet. Interestingly however, unlike the DL models, the ML models (all 6 of them) did not have perfet recall. This indicates that the selected ML models have architectures which make them less supseptible to class imbalances, something which was not taken into account during the implementation of the DL models.

However, the excellent performance of the models may (at least in part) actually indicate that these datasets may have gathered the data for their positive and negative classes with a methodology that unknowingly introduced some level of data leakage. When experimenting with the LIME implementation on the BERT model trained on WELFake, it was found that the model attributed part of its reasoning to news agency names like "Reuters". This is counterproductive to any model developed for fake news detection as models learn to classify news based on superficial attributes instead of ’deeper’ or more substantive attributes.

Moving on, regarding the unique performance of models on the FakeNewsNet dataset, it is worth noting it differs from the other datasets as it has a significant class imbalance, with about 75% of the records belonging to the positive class. Furthermore, FakeNewsNet (in our implementation of the dataset, as no attempt was made to scrape the article body) only contains the title of the article, and does not contain the main article body. For these reasons, FakeNewsNet presents a unique challenge which may explain the relatively (compared to other datasets) poor performance of BERT.

6.2 Cross-Dataset Evaluation

As for cross-dataset evaluation, the performance of the ML and DL models are significantly poorer compared to BERT's cross-dataset evaluation results. With the differences being most notable when looking at the results for the FakeNewsNet and Fake News Detection datasets. The difference in F1-score for FakeNewsNet and Fake News Classification cross-dataset evaluation results between BERT and the CNN model was 0.1649 and 0.2184 respectively.

As for the cross-dataset evaluation performance on the models trained on WELFake. The results for the models trained (regardless if it were BERT, DL or ML) followed a pattern: very strong performance on ISOT, decent performance for FakeNewsNet, poor performance for Fake News Detection (except for the ML models, for which Fake News Detection results were similar to Fake News Classification) and abysmal performance on Fake News Classification.

The models most likely performed better on the FakeNewsNet and ISOT datsets due to the similarities between WELFake and the aforementioned datasets. WELFake contains the article title (originally in its own column; appended at the beginning with the rest of the text during preprocessing) which may be why it performs well on the FakeNewsNet dataset. As for ISOT, it may be that these two datasets are very similar, and as a result the model was able to generalise well for ISOT.

And for the last two datasets: Fake News Detection and Fake News Classification, these datasets likely performed poorly due to being too different than WELFake. As a result the model was unable to generalise for these models thus causing the poor results. This is despite both datsets having 'title' columns initially. This may be because they also contained a 'text' column, which contained the rest of the article; this is in contrast to FakeNewsNet, which contained only the 'title' column. Furthermore, the methodology for collecting these datasets may have been substantially different than the methodology used to collect the WELFake dataset, causing the datasets to differ substantially.

6.3 TPA-BERT

Addressing the proposed model (TPA-BERT) being unable to generalise better than the BERT model, there could be several causes for this. Namely, lack of depth to the text perturbation technique, use of the CLS token, and the fact these perturbations are not 'natural'.

The text perturbation techniques employed were: synonym substitution, random swapping and random deletion. These techniques most likely were unable to meaningfully change the overall semantic meaning of the textual input. This flows directly to the next point, use of the CLS token: the CLS token, as previously mentioned is a 768 dimensional representation of the input text (essentially a semantic summary of the input text). This CLS token makes the order of the text less relevant, as the classifier only uses this CLS token. This means that without genuine semantic variation, the adversarial training only taught the model how to 'autocorrect' the perturbations in the text.

Furthermore, the perturbations do not produce meaningful variations. The random synonym substitution, swapping and deletion creates text which is not a meaningful variation of human writing. This means that while the model learned to adapt to the perturbations, it didn't learn to actually handle genuine variations in human writing styles. Had there been geuine variations, the semantics may have changed as well, allowing the adversarial training to force the model to identify variation-agnostic features while teaching the model to classify the 'fakeness' of news based on less superficial features.

6.4 LIME

Lastly, LIME was implemented to explain the model's (BERT and TPA-BERT) behaviour. The evaluation of the average attribution strength and faifthfulness showed that LIME was able to explain the results of BERT and TPA-BERT to varying degrees, being much more suitable for explaining the BERT model than TPA-BERT. Going into more depth regarding the difference in the results between TPA-BERT and BERT, it does seem that TPA-BERT was sigificantly less explainable than BERT. However there are a few reasons why could have caused this. Namely, the adversarial training process used to train TPA-BERT.

TPA-BERT's adversarial training process trained the model to effectively adapt to missing words and jumbled text, which coincidentally was how LIME measured the attribution. This adversarial training likely made the model more resilient to words being dropped and as such is responsible for the poor attribution and faifthfulness results relative to BERT. Although this research only conducted dataset perturbation, it is possible that adversarial training may indeed reduce the explainability of models, at least when it comes to implementing LIME. However, while these results are not exactly positive for TPA-BERT, it does prove that TPA-BERT does have different behaviour compared to BERT, and that the adversarial training process worked (although not as intended).

7 Conclusion

In summary, this research has, pursuant to the research objectives: completed a thorough literature review of related works, gathered datasets and conducted experiments involving the proposed model (TPA-BERT), BERT and Word2Vec. BERT and TPA-BERT were then fitted with a LIME component for explainability.

For the literature review, 16 articles were reviewed, through which substantial insight was gained into the research around fake news detection. Insight was gained into the use of various embedding algorithms such as BERT, GloVe and Doc2Vec for text, as well as other techniques such as VGG19 for image processing. Furthermore, the various approaches undertaken by the authors of the read papers were noted, and informed the development of the proposed model for this research. These approaches varied from machine learning, to multi-layered models employing embedding algorithms, deep learning and graphical models amongst other algorithms. Most interestingly, many state-of-the-art models were black-boxes, which motivated the reading of articles discussing explainable AI techniques, such as LIME. To conclude, the literature review was critical to the development of a substantial knowledge base which supported this FYP and efforts toward the research goals.

Relevant datasets were gathered from Kaggle, research papers and HuggingFace. Originally seven datasets were gathered, but two of them proved to be incompatible and were not used. The five chosen datasets were previously described in detail in Chapter 3. These datasets were then cleaned and preprocessed, allowing for their use in the later experiments with the various embedding algorithms.

Experiments on BERT and Word2Vec were conducted which aimed to gain baseline results for comparison against the proposed model, and to justify the use of BERT compared to other techniques. The experimental data showed that while models (both ML and DL) trained on Word2Vec embeddings performed marginally worse than BERT, the real difference lay in generalisation, where BERT models outperformed the models trained on Woed2Vec embeddings. While this showed BERT had superior generalisation, the patterns observed in the cross-dataset evaluation results suggested that similarity between the dataset used to train a model and the completely unseen datasets used for cross-dataset evaluation is an important factor.

The proposed model was constructed based on approaches to fake news detection in the articles reviewed in the literature review. It had been hoped that the proposed model would have greater generalisation than baseline BERT models, however this result was not obtained. Rather the results indicate that TPA-BERT was no better or worse than the baseline BERT models in terms of performance (against the test set of the dataset it was trained on) and cross-dataset evaluation. It can be concluded, at least for the datasets involved in this research, that merely perturbing the text using the aforementioned techniques may be too 'shallow' a technique for effective generalisation.

And finally, LIME was implemented on BERT and TPA-BERT with results indicating that LIME had greater explainability for BERT than TPA-BERT. This was likely due to the adversarial training process employed by TPA-BERT which made it resilient to words being randomly removed. Despite this, these results prove that TPA-BERT's adversarial training process did impact the model's behaviour, just not as intended (increased generalisation).

7.1 Limitations

This research contains several limitations, such as lack of hyperparameter tuning, large batch sizes, limited LIME implementation.

Hyperparameter tuning was not conducted due to limitations in computational resources. It is possible with sufficient hyperparameter tuning, TPA-BERT could've performed better.

Furthermore, the limitation in computational resources was also the rationale behind the use of large batch sizes in the training parameters of the BERT and TPA-BERT models. This could have led to a deprecation in performance both in regards to same-dataset performance and cross-dataset performance.

Lastly, LIME was only implemented and evaluated on the BERT and TPA-BERT models trained on WELFake. This was done primarily due to the intense computational demands of LIME. In addition, in the implementation of LIME, each input was only explained by 1000 perturbed samples (again, due to resource constraints). An increase of this number to 5000 wold ahve likely improved explainability.

8 Future Work

Future work could expand on this research by the use of 'deeper' adversarial methods, more diverse datasets, BERT varieties as well as exploring multimodal methods.

Firstly, hyperparameter optimisation, as mentioned in the limitations section, could have improved model performance. Future work could employ hyperparemter optimisation to improve model performance and also explore the impact of specific parameters on generalisation.

Secondly, more datasets or more specifically, more diverse datasets could be employed. Although not done in this research, combination of datasets will likely improve model generalisation by exposing the model to multiple 'domains' (datasets) during training. And should more datasets be procured there is also more evaluation data available for a more thorough analysis.

Furthermore, in this research only BERT (TPA-BERT also used BERT) was used. Future work could explore the use of BERT varieties such as RoBERTa, DistilBERT, or ELECTRA (Efficiently Learning an Encoder that Classifies Token Replacements Accurately) to name but a few. Some BERT variants may have superior generalisation performance in regards to fake news detection.

And lastly, this research only explored text-based methods, future work could explore the use of multimodal methodologies for fake news detection. Throughout the literature review, multimodal techniques generally had the input structure as an image (accompanying the text) and a body of text. A possible alternative technique for future work could see the input structure as: an image of the webpage containing the article, and the text of the article. Such a methodology may lead to greater rates of fake news detection and improved generalisation as there may be stylistic differences between sites more likely to publish genuine news and sites which predominantly publish fake news.

References

Ahmed, H., Traore, I., & Saad, S. (2017). Detection of online fake news using n-gram analysis and machine learning techniques. *International conference on intelligent, secure, and dependable systems in distributed and cloud environments*, 127–138.

Ahmed, H., Traore, I., & Saad, S. (2018). Detecting opinion spams and fake news using text classification. *Security and Privacy*, *1*(1), e9.

Ahmed, M., Hossain, M. S., Islam, R. U., & Andersson, K. (2022). Explainable text classification model for covid-19 fake news detection. *J. Internet Serv. Inf. Secur.*, *12*(2), 51–69.

Ali, A. M., Ghaleb, F. A., Al-Rimy, B. A. S., Alsolami, F. J., & Khan, A. I. (2022). Deep ensemble fake news detection model using sequential deep learning technique. *Sensors*, *22*(18). https://doi.org/10.3390/s22186970

Baptista, J. P., & Gradim, A. (2022). A working definition of fake news. *Encyclopedia*, *2*(1), 632–645. https://doi.org/10.3390/encyclopedia2010043

Chen, Y.-P., Chen, Y.-Y., Yang, K.-C., Lai, F., Huang, C.-H., Chen, Y.-N., & Tu, Y.-C. (2022). The prevalence and impact of fake news on covid-19 vaccination in taiwan: Retrospective study of digital media. *Journal of Medical Internet Research*, *24*(4). https://doi.org/10.2196/36830

Chien, S.-Y., Yang, C.-J., & Yu, F. (2022). Xflag: Explainable fake news detection model on social media. *International Journal of Human–Computer Interaction*, *38*(18-20), 1808–1827. https://doi.org/10.1080/10447318.2022.2062113

Dhiman, P., Kaur, A., Gupta, D., Juneja, S., Nauman, A., & Muhammad, G. (2024). Gbert: A hybrid deep learning model based on gpt-bert for fake news detection. *Heliyon*, *10*(16).

E.Almandouh, M., Alrahmawy, M. F., Eisa, M., Elhoseny, M., & Tolba, A. S. (2024). Ensemble based high performance deep learning models for fake news detection. *Scientific Reports*, *14*(1). https://doi.org/10.1038/s41598-024-76286-0

Hashmi, E., Yayilgan, S. Y., Yamin, M. M., Ali, S., & Abomhara, M. (2024). Advancing fake news detection: Hybrid deep learning with fasttext and explainable ai. *IEEE Access*, *12*, 44462–44480. https://doi.org/10.1109/ACCESS.2024.3381038

Jing, J., Wu, H., Sun, J., Fang, X., & Zhang, H. (2023). Multimodal fake news detection via progressive fusion networks. *Information Processing & Management*, *60*(1), 103120. https://doi.org/https://doi.org/10.1016/j.ipm.2022.103120

Khanam, Z., Alwasel, B. N., Sirafi, H., & Rashid, M. (2021). Fake news detection using machine learning approaches. *IOP Conference Series: Materials Science and Engineering*, *1099*(1), 012040. https://doi.org/10.1088/1757-899x/1099/1/012040

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). Roberta: A robustly optimized bert pretraining approach. https://arxiv.org/abs/1907.11692

Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems*, *26*.

Qin, S., & Zhang, M. (2024). Boosting generalization of fine-tuning bert for fake news detection. *Information Processing & Management*, *61*(4), 103745. https://doi.org/https://doi.org/10.1016/j.ipm.2024.103745

Saleh, H., Alharbi, A., & Alsamhi, S. H. (2021). Opcnn-fake: Optimized convolutional neural network for fake news detection. *IEEE Access*, *9*, 129471–129489. https://doi.org/10.1109/ACCESS.2021.3112806

Savage, M. (2025, December). Youtube channels spreading fake, anti-labour videos viewed 1.2bn times in 2025. https://www.theguardian.com/technology/2025/dec/13/fake-anti-labour-video-billion-views-youtube-2025

Shu, K., Mahudeswaran, D., Wang, S., Lee, D., & Liu, H. (2018). Fakenewsnet: A data repository with news content, social context and dynamic information for studying fake news on social media. *arXiv preprint arXiv:1809.01286*.

Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake news detection on social media: A data mining perspective. *ACM SIGKDD Explorations Newsletter*, *19*(1), 22–36.

Shu, K., Wang, S., & Liu, H. (2017). Exploiting tri-relationship for fake news detection. *arXiv preprint arXiv:1712.07709*.

Szczepański, M., Pawlicki, M., Kozik, R., & Choraś, M. (2021). New explainability method for bert-based model in fake news detection. *Scientific Reports*, *11*(1). https://doi.org/10.1038/s41598-021-03100-6

Wu, J., Guo, J., & Hooi, B. (2024). Fake news in sheep's clothing: Robust fake news detection against llm-empowered style attacks. *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 3367–3378. https://doi.org/10.1145/3637528.3671977

Xue, J., Wang, Y., Tian, Y., Li, Y., Shi, L., & Wei, L. (2021). Detecting fake news by exploring the consistency of multimodal data. *Information Processing & Management*, *58*(5), 102610. https://doi.org/https://doi.org/10.1016/j.ipm.2021.102610

Yuan, H., Zheng, J., Ye, Q., Qian, Y., & Zhang, Y. (2021). Improving fake news detection with domain-adversarial and graph-attention neural network. *Decision Support Systems*, *151*, 113633. https://doi.org/https://doi.org/10.1016/j.dss.2021.113633

Zhang, Q., Guo, Z., Zhu, Y., Vijayakumar, P., Castiglione, A., & Gupta, B. B. (2023). A deep learning-based fast fake news detection model for cyber-physical social services. *Pattern Recognition Letters*, *168*, 31–38. https://doi.org/https://doi.org/10.1016/j.patrec.2023.02.026

Zhang, Z., Lv, Q., Jia, X., Yun, W., Miao, G., Mao, Z., & Wu, G. (2024). Gbca: Graph convolution network and bert combined with co-attention for fake news detection. *Pattern Recognition Letters*, *180*, 26–32. https://doi.org/https://doi.org/10.1016/j.patrec.2024.02.014

Appendix
Meeting Logs

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 7th November 2025	**Meeting No.: 1**
Meeting Mode: Online	
Project ID: **351**	**Project Type:** Research-based
Project Title: **Fake news detection using deep learning**	
Student ID: **241UC24151**	**Student Name:** **Jordan Ling Shen Hang**
Student Programme and Specialisation: **Bachelor of Computer Science (Data Science)**	
Supervisor Name: **Assoc. Prof. Dr. Chua Sook Ling**	**Co-Supervisor Name:** **(if applicable)** **Dr. Foo Lee Kien**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE

Tasks: Problem Formulation and Project Planning

Details (in point form):

[First meeting, so no tasks completed before the meeting]

- **Briefed about the structure of the FYP**
- **Project tasks for the next two semesters laid out**
- **Given tasks (introduction, literature review reading) to start the project**

2. WORK TO BE DONE

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review

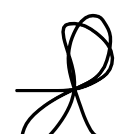
Details (in point form):

- **Write problem statement, research objectives and scope (Chapter 1)**
- **Begin reading and writing for the literature review**

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

None due to first meeting

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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Supervisor's Signature

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Co-Supervisor's Signature
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CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 21st November 2025	**Meeting No.: 2**
Meeting Mode: Online	
Project ID: **351**	**Project Type:** Research-based
Project Title: **Fake news detection using deep learning**	
Student ID: **241UC24151**	**Student Name:** **Jordan Ling Shen Hang**
Student Programme and Specialisation: **Bachelor of Computer Science (Data Science)**	
Supervisor Name: **Assoc. Prof. Dr. Chua Sook Ling**	**Co-Supervisor Name:** **(if applicable)** **Dr. Foo Lee Kien**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review

Details (in point form):

- Worked on problem statement, research scope and research objectives (Chapter 1)
- Read and wrote about 6 articles for the literature review

2. WORK TO BE DONE

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Prototype Development or Proof of Concept

Details (in point form):

- Continue working on literature review
- Revamp problem statement (a bit lacking in background, and a bit lacking overall)
- Begin working with and analysing datasets

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

Was unsure if literature review was written properly; solution was that I sent my draft report to my supervisor for proofreading.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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Company Supervisor's Signature
(if applicable)

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CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 2nd December 2025	**Meeting No.: 3**
Meeting Mode: Online	
Project ID: 351	**Project Type:** Research-based
Project Title: **Fake news detection using deep learning**	
Student ID: **241UC24151**	**Student Name:** **Jordan Ling Shen Hang**
Student Programme and Specialisation: **Bachelor of Computer Science (Data Science)**	
Supervisor Name: **Assoc. Prof. Dr. Chua Sook Ling**	**Co-Supervisor Name:** **(if applicable)** **Dr. Foo Lee Kien**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE

Tasks: Background Study or Literature Review / Draft Report Completion

Details (in point form):
- More articles for literature review
- Some work on problem statement

2. WORK TO BE DONE

Tasks: Background Study or Literature Review / Prototype Development or Proof of Concept/ Draft Report Completion

Details (in point form):
- Work on analysing datasets
- More articles for literature review
- Fix citations (update to APA7)
- Remove unneeded figures

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

LaTeX APA formatting is outdated; researched and found a way to apply APA7 to LaTeX.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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(if applicable)

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Company Supervisor's Signature
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CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 16th December 2025	**Meeting No.: 4**
Meeting Mode: Online	
Project ID: **351**	**Project Type:** Research-based
Project Title: **Fake news detection using deep learning**	
Student ID: **241UC24151**	**Student Name:** **Jordan Ling Shen Hang**
Student Programme and Specialisation: **Bachelor of Computer Science (Data Science)**	
Supervisor Name: **Assoc. Prof. Dr. Chua Sook Ling**	**Co-Supervisor Name:** **(if applicable)** **Dr. Foo Lee Kien**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE

Tasks Study or Literature Review / Prototype Development or Proof of Concept / Draft Report Completion

Details (in point form):
- Worked on analysing datasets (only one so far)
- Fixed citations in latex (update to APA7)
- Read and reviewed more articles for the literature review
- Removed unneeded figures

2. WORK TO BE DONE

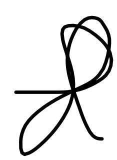
Tasks: Prototype Development or Proof of Concept / Draft Report Completion

Details (in point form):
- Work on experimenting with the dataset.
- Document analysis in the report

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

None

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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Co-Supervisor's Signature
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Company Supervisor's Signature
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FACULTY OF COMPUTING & INFORMATICS

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CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 29th December 2025	**Meeting No.: 5**
Meeting Mode: Online	
Project ID: **351**	**Project Type:** Research-based
Project Title: **Fake news detection using deep learning**	
Student ID: **241UC24151**	**Student Name:** **Jordan Ling Shen Hang**
Student Programme and Specialisation: **Bachelor of Computer Science (Data Science)**	
Supervisor Name: **Assoc. Prof. Dr. Chua Sook Ling**	**Co-Supervisor Name:** **(if applicable)** **Dr. Foo Lee Kien**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE

Tasks: Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

Details (in point form):

- Cleaning and processing of datasets
- Some basic EDA on the datasets
- Documented datasets in report
- Experimentation with GloVe and Word2Vec
- Some progress on fine-tuning BERT

2. WORK TO BE DONE

Tasks: Prototype Development or Proof of Concept / Draft Report Completion

Details (in point form):

- Continue experiments with BERT
- Document results

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

- Encountered an issue with having limited computational power on my local laptop. Plan to use Google Colab to resolve this limitation.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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Company Supervisor's Signature
(if applicable)

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 20th January 2026	**Meeting No.: 6**
Meeting Mode: Online	
Project ID: 351	**Project Type:** Research-based
Project Title: **Fake news detection using deep learning**	
Student ID: **241UC24151**	**Student Name:** **Jordan Ling Shen Hang**
Student Programme and Specialisation: **Bachelor of Computer Science (Data Science)**	
Supervisor Name: **Assoc. Prof. Dr. Chua Sook Ling**	**Co-Supervisor Name:** **(if applicable)** **Dr. Foo Lee Kien**
Collaborating Company: **(if applicable)**	**Company Supervisor Name:** **(if applicable)**

1. WORK DONE

Tasks: Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

Details (in point form):

- Completed experimentation with GloVe, Word2Vec and BERT.
- Documented results in report
- Partial work done on discussion of results
- Worked on documenting the methodology

2. WORK TO BE DONE

Tasks: Prototype Development or Proof of Concept/ Draft Report Completion

Details (in point form):

- Writing of the theoretical framework chapter
- Complete results discussion
- Refining report and formatting

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

None

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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Company Supervisor's Signature
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Appendix
Turnitin Similarity

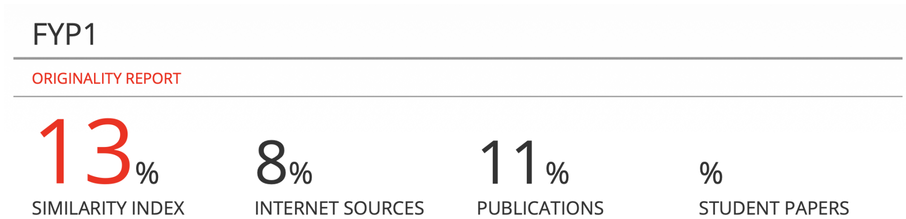

Figure B1: Turnitin Similarity Score for This Report